



Antibiotic resistance genes prediction via whole genome sequence analysis of *Stenotrophomonas maltophilia*

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ABSTRACT

Background: *Stenotrophomonas maltophilia* (*S. maltophilia*) is the first dominant ubiquitous bacterial species identified from the genus *Stenotrophomonas* in 1943 from a human source. *S. maltophilia* clinical strains are resistance to several therapies, this study is designed to investigate the whole genome sequence and antimicrobial resistance genes prediction in *Stenotrophomonas maltophilia* (*S. maltophilia*) SARC-5 and SARC-6 strains, isolated from the nasopharyngeal samples of an immunocompromised patient.

Methods: These bacterial strains were obtained from Pakistan Institute of Medical Sciences (PIMS) Hospital, Pakistan. The bacterial genome was sequenced using a whole-genome shotgun via a commercial service that used an NGS (Next Generation Sequencing) technology called as Illumina HiSeq 2000 system for genomic sequencing. Moreover, detailed *in-silico* analyses were done to predict the presence of antibiotic resistance genes in *S. maltophilia*.

Results: Results showed that *S. maltophilia* is a rare gram negative, rod-shaped, non sporulating bacteria. The genome assembly results in 24 contigs (>500 bp) having a size of 4668,850 bp with 65.8% GC contents. Phylogenetic analysis showed that SARC-5 and SARC-6 were closely related to *S. maltophilia* B111, *S. maltophilia* BAB-5317, *S. maltophilia* AHL, *S. maltophilia* BAB-5307, *S. maltophilia* RD-AZPVI_04, *S. maltophilia* JF22, *S. maltophilia* RD_MAAMIB_06 and lastly with *S. maltophilia* sp ROI7. Moreover, the whole genome sequence analysis of both SARC-5 and SARC-6 revealed the presence of four resistance genes *adeF*, *qacG*, *adeF*, and *smeR*.

Conclusion: Our study confirmed that *S. maltophilia* SARC-5 and SARC-6 are one of the leading causes of nosocomial infection which carry multiple antibiotic resistance genes.

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Introduction

The genus *Stenotrophomonas* belongs to the phylum *pseudomonadota* and the class *gammaproteobacteria* which consists of total 18

validated species in the list of Prokaryotic Names Standing in Nomenclature. The members of this genus commonly occurred in diverse habitats including soil, plant-associated reservoirs, tap water, soda lakes, sewage, eroded metal surfaces and animal and human bodies [1]. *Stenotrophomonas maltophilia* (*S. maltophilia*) is the first dominant ubiquitous bacterial species identified from the genus *Stenotrophomonas* in 1943 from a human source. This species was classified as *Bacterium booker*, reclassified as *Pseudomonas maltophilia* and transferred to the genus *Xanthomonas* as

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Xanthomonas maltophilia [2]. All other 11 species of the genus *Stenotrophomonas* were obtained from different environmental sources which include *Stenotrophomonas rhizophila* (plant source) [3], *Stenotrophomonas dokdonensis* (soil source) [4], *Stenotrophomonas koreensis* (animal compost) [5], *Stenotrophomonas humi* (soil source) [6], *Stenotrophomonas terrae* (soil source) [7], *Stenotrophomonas chelati-phaga* (sewage) [8], *Stenotrophomonas ginsengisoli* (soil source) [9], *Stenotrophomonas daejeonensis* (sewage) [10], *Stenotrophomonas pavanii* (plant source) [11], *Stenotrophomonas tumulicola* (spots and gels) [12] and *Stenotrophomonas DDT-1* (contaminated environment) [13]. Furthermore, there are many nomenclatural alterations occurred in this genus [14]. At first, *Stenotrophomonas africana* was considered a novel species of this genus but later on, it was proposed as similar to *S. maltophilia* on the basis of DNA-DNA hybridization analysis [15]. *Stenotrophomonas dokdonensis* was initially assigned to the genus *Pseudoxanthomonas* and converted to the genus *Stenotrophomonas* afterwards [16]. Moreover, three different species *Pseudomonas beteli* (4.48Mbp), *Pseudomonas hibiscicola* (4.42Mbp) and *Pseudomonas geniculata* (4.80Mbp) were first kept under the genus *Pseudomonas* and transferred to genus *Stenotrophomonas* and has similar properties as that of *S. maltophilia* [17]. The members of the *Stenotrophomonas* genus have metabolic adaptability and intraspecific diversification with numerous biotechnological applications [18]. Distinct *Stenotrophomonas* species are capable to generate antimicrobial compounds that can protect plants from disease damage and enhance the growth of plants by producing plant growth promoting factors. These species can enhance the overall plant yield due to the increased production of indole-3 acetic acid (IAA) [19]. Some *Stenotrophomonas spp.* have the ability to show resistance against toxic heavy metals and are significantly used in bioremediation [20]. Members of this genus potentially participate in nitrogen fixation, sulphur oxidation and promote the symbiotic association with different plant species. Most of the *Stenotrophomonas* species are habitual to the nutrient restricted environment but can colonize and adhere to the plant surfaces by producing pilli and form complex biofilms [21]. Due to the property of biofilm formation and adhesion potential, *Stenotrophomonas* members may compete with other microbes on the surface of plant roots and maintain their survival for a longer run [22].

Apart from the symbiotic associations of different members of this genus with plants, some species, particularly *S. maltophilia* showed adverse effects when exposed to human hosts, specifically in hospital settings. *S. maltophilia* is a rod-shaped, gram-negative, non-fermenting bacteria that is responsible for causing nosocomial infections such as bacteremia, endocarditis, respiratory tract ailments, and urinary tract infections in human individuals [23]. It is considered as a pathogenic entity among immunocompromised patients who are already under the influence of numerous co-morbidities. According to the epidemiological analysis, it is present in China [24], the United States [25], Germany [26], Canada [27], Iran [28], and Bangladesh [29]. Hence its occurrence could not be confined to a particular region. This huge covering of the globe indicates that it is a matter of public health concern and should be given proper attention in order to avoid the potential losses of human lives.

There have been various antibiotics that can be used for the treatment of infection caused by *S. maltophilia*. The treatment options for *S. maltophilia* may include trimethoprim-sulfamethoxazole, ticarcillin-clavulanic acid, minocycline, fluoroquinolones, and tigecycline [30]. Furthermore, the strains of *S. maltophilia* remain susceptible to colistin, trimethoprim-sulfamethoxazole, netilmicin, and ciprofloxacin [31]. A total of 5.87% of the *S. maltophilia* strains were extensively drug-resistant with about 12.1% of strains being resistant to trimethoprim-sulfamethoxazole and 8.99% of isolates being resistant to levofloxacin. Moreover, this bacterium is also resistant to meropenem, imipenem, and doripenem [32].

The antibiotic resistance in *S. maltophilia* is due to the production of antibiotic-degrading enzymes like extended-spectrum beta-lactamases [33], Amp C type beta-lactamases [34], beta-lactamase enzymes L1 and L2, loss of outer membrane proteins and the presence of efflux pumps [35]. The resistance patterns observed in *S. maltophilia* are due to its genomic alternation. The genes responsible for the trimethoprim-sulfamethoxazole resistance are the dihydropteroate synthase *sul1* and *sul2* as well as dihydrofolate reductase *dhfrA* [36]. A total of 62.8% of isolates exhibited *smqnr* gene which has the potential to confer resistance against fluorquinolones [32]. Furthermore, these isolates also carried other genes including; *blaL1*, *blaL2*, *aph(3')IIC*, *sul1*, and *sul2*. The *smeD* mediated efflux system is also reported to be prevalent in the *S. maltophilia* isolates that contribute to the resistance against trimethoprim-sulfamethoxazole. *S. maltophilia* also exhibited biofilm formation genes including; *rpff*, *spgM*, and *rmlA*. These genes indicate a further increase in antibiotic resistance, as biofilms are considered one of the most important factors for increased antibiotic resistance. Moreover, *S. maltophilia* also acquired resistance genes like *qnrA*, *qnrB*, *oqxA*, *oqxB* and *qnrS* [37].

Antimicrobial resistance (AMR) continues to be a major public health concern in Pakistan. *S. maltophilia* is found in around 6.2% of the Pakistani population, ranging in age from < 5 to > 65. About 20.3% were detected in the < 5 age group, 1.15% in the 6–18 age group, 1.1% in the 19–45 age group, 0.55% in the 46–65 age group, and 1.4% in the > 65 age group. It affects approximately 6.1% of Pakistani males and 8.8% of Pakistani females. Therefore, it is critical to understand the genetic basis of this bacterium in the Pakistani population. This study is designed to explore the whole genomic analysis of *Stenotrophomonas maltophilia* species and characteristic antimicrobial resistance profiling from hospitalized immunocompromised patients.

Materials and methods

Materials

Cotton swab, Nutrient Agar Plates, Incubator, Microscope, Centrifuge, DNA isolation kit, Polymerase Chain Reaction (PCR), Nanodrop, and other materials were used in this study. All chemicals, reagents, and other consumables were analytical grade and procured from a standard chemical supplier.

Bacterial isolation

The bacterial strains SARC-5 and SARC-6 were isolated from the nasopharyngeal sample of an immune-deficient patient at the Hospital of Pakistan Institute of Medical Sciences (PIMS), Islamabad, Pakistan. This study includes total 24 samples, 12 male and 12 female with age above 40 and the samples were collected from individuals voluntarily. On 5% blood agar plates, the microbes were grown and kept at $35 \pm 2^\circ \text{C}$ for 2 days. The strains were initially identified by biochemical assays and morphological observations. On blood agar plates, the expected colonies of SARC-5 and SARC-6 were cultivated. All of the protocols were approved by the Ethical Committee of PIMS Hospital.

Molecular identification of isolates

DNA extraction

Pure isolates were grown on blood agar media and then incubated at 30°C for 24 h to get the fresh culture. For the extraction of genomic DNA, Genomic DNA kit (Thermo Fisher) method was used. Initially, microbial pellets were placed in a tube and sodium dodecyl sulfate (SDS) and proteinase K were added for enzymatic digestion of protein and cellular components. After this, the mixture of phenol:

chloroform: isoamyl alcohol (25:24:1) was prepared to separate DNA from cellular debris. Centrifugation at 13,000 rpm for 5 min was done to obtain the aqueous phase which contains DNA. After initial centrifugation, 0.6% of isopropanol was added and the tube was centrifuged again to collect the DNA pellet. The quality and quantity of the obtained DNA were examined using NanoDrop 2000c spectrophotometer (Thermo Scientific, USA) at 260 and 280 nm optical wavelengths.

Genomic sequencing and analysis

The bacterial genome was sequenced using a whole-genome shotgun via a commercial service. DNA library was prepared with Covaris 220 and sequencing was performed with Illumina HiSeq 2000 platform (2×150-bp reads). The genomic data were analyzed using the online RAST server. The bacterial whole genomic sequence quality was evaluated by using the FastQC. FastQC, is a freely available bioinformatics tool used to provide all information related to basic quality control metrics for raw whole genomic sequencing data [38]. Sequencer produced 11636845 and 253446 pair end reads with 66% and 64% GC content of SARC-5 and SARC-6 respectively. The average sequence length of both reads was 150 bp and no sequence was flagged as poor quality (QC files attached at the end). Genome based molecular identification was conducted with KmerFINDER and genome assembly was generated via <https://github.com/rwrick/Unicycler>. A circular genome map was generated using online CGviewer (<http://cgview.ca/>) and nucleotide skew was calculated by using GenSkew (<https://genskew.csb.univie.ac.at/>). Antimicrobial resistance genes and virulence factors were determined using comprehensive antibiotics resistance database (CARD) (<https://card.mcmaster.ca/>) and VirulenceFinder (<https://cge.cbs.dtu.dk/services/VirulenceFinder/>), respectively. The pathogenicity of the strain was predicted by using PathogenFinder 1.1 (<https://cge.cbs.dtu.dk/services/PathogenFinder/>). Moreover, comparative proteome analyses were performed and the Venn diagram was generated using Ortho Venn [39]. Further comparative analysis of genomic composition and resistance genes was performed with *S. maltophilia* from China population (Accession Number: CP071784). For consistency the genome was re-annotated using RAST and resistant genes was searched using CARD.

Results

S. maltophilia is a rare gram negative, rod-shaped, non-sporulating bacteria (Fig. 1). Classification and characteristic features of *S. maltophilia* are mentioned in Table 1. Moreover, morphological and biochemical identification of isolates are depicted in Table 2.

Whole genome testing of *S. maltophilia*

S. maltophilia is an opportunistic bacteria isolated from immunocompromised patients. Genome assembly results show that this bacterium consists of a total 4,668,850 bp genome with 65.8% GC content and contains 4325 coding sequences with 71 tRNAs. RAST functional annotation shows that the genome of *S. maltophilia* contains 190 genes for protein, 67 for fatty acid, lipid and isoprenoids, 15 for nitrogen, 27 for phosphorus, 150 for carbohydrates, 9 for potassium and 3 for sulfur metabolism. Genes involved in structural regulation are 25 for cell wall and capsule, 5 for cell division and cell cycle, 18 for motility and chemotaxis and 162 for membrane transport. Furthermore, total genes for virulence, disease and defense are 37 and 61 for environmental stress response (Fig. 2). Additionally, *S. maltophilia* from China population consists of 4,391,471 bp genome with 66.5% GC content and contains 4038 coding sequences with 76 tRNAs. RAST annotation showed that it has 195 genes for proteins, 66 for fatty acid, lipid and isoprenoids, 15 for nitrogen, 28 for phosphorus, 160 for carbohydrates, 9 for potassium and 3 for sulfur

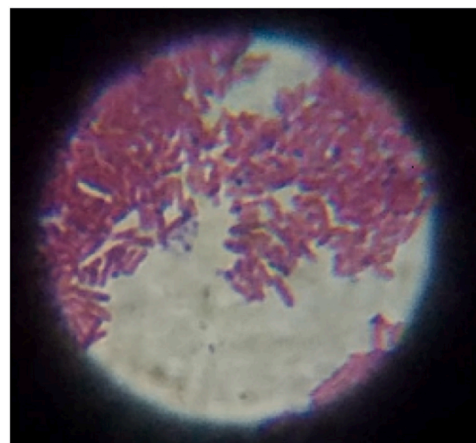


Fig. 1. : Gram-negative rods of *S. maltophilia* isolated from the nasopharyngeal sample of immunocompromised patients.

Table 1
Classification and features of *S. maltophilia*.

Classification	Domain: <i>Bacteria</i> Phylum: <i>Pseudomonadota</i> Class: <i>Gammaproteobacteria</i> Order: <i>Xanthomonadales</i> Family: <i>Xanthomonadaceae</i> Genus: <i>Stenotrophomonas</i> Species: <i>Stenotrophomonas maltophilia</i>
Habitat	Plants, soils, raw milk and human skin
Pathogenicity	Opportunistic pathogen
Gram stain	Negative
Colony	Small, circular, slightly raised
Motility	Motile
Sporulation	Non spore forming
Oxygen consumption	Aerobic
Temperature range	Mesophilic (35± 2°C)
Lysine decarboxylase	Positive
Catalase	Positive
Oxidase	Negative
Aesculin hydrolysis	Positive
Gelatin liquefaction	Positive

Table 2
Morphological and biochemical identification of isolates.

Isolates	<i>S. maltophilia</i> SPAC-5	<i>S. maltophilia</i> SPAC-6
Colony surface	Small, circular and raised	Small, circular and raised
Colony size	0.4– 0.7µm	0.4–0.7µm
Colony color (BA)	Yellow pigmented	Slightly yellow pigmented
Colony color (MA)	Pale color less	Pale color less
Gram staining	Gram negative	Gram negative
Motility	Positive	Positive
Catalase test	Positive	Positive
Oxidase test	Negative	Negative
DNAse test	Positive	Positive
Aesculin hydrolysis	Positive	Positive
Gelatin liquefaction	Positive	Positive
Lysine decarboxylase	Positive	Positive

metabolism. Genes involved in structural regulation are 25 for cell wall and capsule, 3 for cell division and cell cycle, 18 for motility and chemotaxis and 153 for membrane transport. Furthermore, total genes for virulence, disease and defense are 31 and 60 for environmental stress response. The circular graph of CGView shows GC content present unevenly throughout the genome at different positions. In GC skew, asymmetrical distribution of replicons is observed in the bacterial genome. In the leading strand, G over C, and



Fig. 2. : Graph showing functional genomic statistic of *S. maltophilia*.

in the lagging strand, C over G show two replicons regions (Fig. 3). The genomic statistics of *S. maltophilia* are depicted in Table 3.

Phylogenetic analysis of *S. maltophilia*

The phylogenetic tree was constructed by NCBI blast (basic local alignment search tool). This analysis involved sequencing on the basis of 16 s rRNA from six available species of *S. maltophilia*. *S. maltophilia* SARC-5 has 99.60% similarity index with *S. maltophilia* B111 followed by *S. maltophilia* SRC-6 99.38%, *S. maltophilia* BAB-5317 99.31%, *S. maltophilia* AHL, *S. maltophilia* BAB-5307 and *S. maltophilia* RD-AZPVI_04 have 99.21%, 99.09% with *S. maltophilia* JFZ2, 90.02% with *S. maltophilia* RD_MAAMIB_06 and lastly with *S. maltophilia* spp ROi7 98.82%. *S. maltophilia* SARC-6 has similarity with *S. maltophilia* B111 followed by *S. maltophilia* AHL, *S. maltophilia* BAB-5307, *S. maltophilia* 5317, *S. maltophilia* JFZ2 and *S. maltophilia* RD-AZPVI_04 (Fig. 4).

Table 3
Genomic statistic *S. maltophilia*.

Genome	Specification SARC Pakistani	Specification CP071784 (China)
Domain	Bacteria	Bacteria
Taxonomy	Bacteria; <i>S. maltophilia</i>	Bacteria; <i>S. maltophilia</i>
Size	4668,850 bp	4397,471
GC content	65.8%	66.5
Number of contigs	24	1
Number of subsystems	315	305
Number of coding sequencing	4325	4038
Number of tRNAs	71	76
Number of rRNAs	13	15

Genomic islands and resistance/virulence genes

Resistance genes annotation was performed to visualize the resistance genes in *S. maltophilia*. Red highlighted inner circle shows GC rich region, while blue and orange highlighted regions indicate the predicted island. These results do not show any significant pathogenic island (Fig. 5). The comprehensive antibiotic resistance database (CARD) shows both strains contain four resistance genes *adeF*, *qacG*, *adeF*, and *smeR*. The strain from China also shows three same genes and one unique *APH (9)-Ic* gene (Table 4 and Table 5). Three belong to the resistance nodulation cell division (RND) efflux pump family which can effectively pump antibiotics out of the cell. One gene is against disinfecting and antiseptics. ResFinder annotation results show resistance genes including *Oqx*B, *bla*L1 and *aph (3')-IIc* and there is no significant results for Chinese strain using ResFinder. *S. maltophilia* isolates which carried *bla*L1 and *aph (3')-IIc* are intrinsically resistant to β -lactams and aminoglycosides class of drugs respectively. *Oqx*B is an acquired plasmid mediated resistance gene belonging to the class RND efflux pump family which causes

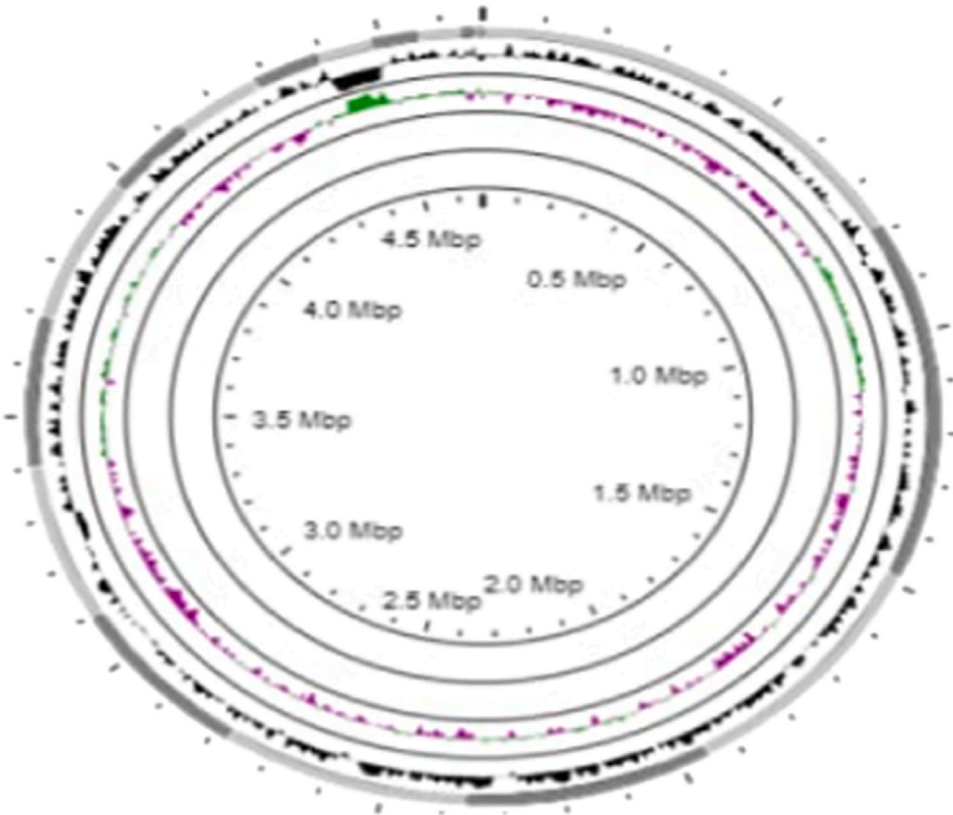
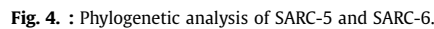


Fig. 3. : Circular maps for chromosomes *S. maltophilia* SARC-5 and SARC-6 generated by CGView.



Orthologous clusters of *Stenotrophomonas* Strains

Comparison of *Stenotrophomonas* SARC-5 and *Stenotrophomonas* SARC-6 with *S. maltophilia*, *pseudomonas aeruginosa* and *Xanthomonas oryzae* shows 4794 clusters, 3547 orthologous clusters

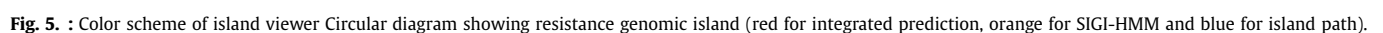


Table 4
Resistance genes annotated on CARD.

RGI Criteria	ARO Term	Detection Criteria	AMR Gene Family	Drug Class	Resistance Mechanism	% Identity of Matching Region	% Length of Reference Sequence
SARC Pakistani	adeF	Protein homolog model	Resistance-nodulation-cell division (RND) antibiotic efflux pump	Fluoroquinolones antibiotic, tetracycline antibiotic	Antibiotic efflux	61.08	99.91
Strict	qacG	Protein homolog model	Small multidrug resistance (SMR) antibiotic efflux pump	Disinfecting agent and antiseptics	Antibiotic efflux	48.08	102.80
Strict	adeF	Protein homolog model	Resistance-nodulation-cell division (RND) antibiotic efflux pump	Fluoroquinolones antibiotic, tetracycline antibiotic	Antibiotic efflux	59.96	99.72
Strict	smeR	Protein homolog model	Resistance-nodulation-cell division (RND) antibiotic efflux pump	aminoglycoside antibiotic, cephalosporin, cephamycin, imipenem	Antibiotic efflux	99.56	100.00
CP071784 (China)	smeR	Protein homolog model	Resistance-nodulation-cell division (RND) antibiotic efflux pump	aminoglycoside antibiotic, cephalosporin, cephamycin, imipenem	Antibiotic efflux	99.56	100.00
Strict	qacG	Protein homolog model	Small multidrug resistance (SMR) antibiotic efflux pump	disinfecting agents and antiseptics	Antibiotic efflux	48.08	102.80
Strict	APH (9)-Ic	Protein homolog model	APH (9)	Aminoglycoside antibiotic	Antibiotic inactivation	98.2	96.53
Strict	adeF	Protein homolog model	Resistance-nodulation-cell division (RND) antibiotic efflux pump	Fluoroquinolone antibiotic, tetracycline antibiotic	Antibiotic efflux	60.06	99.72

Table 5
Resistance/virulence genes of *S. maltophilia*.

Genes	Identity	Coverage	Position in reference	Intrinsic/Acquired
OqxB	76.17	97.62	1.3121	Acquired
blaL1	89.6	91.26	57,852	Intrinsic
aph (3')-IIc	89.11	99.13	1.807	Intrinsic

and 1247 single copy gene clusters. 4147. These five strains have 2254, 3712, 2506, 4147 and 4147 orthologous clusters respectively. SARC-5 and SARC-6 both strains contain a 4179-protein count with 9 singletons (Fig. 6). This analysis shows the similarity index between multiple species. The large group of common proteins is most probably a basic cell structure of *S. maltophilia*. Moreover, comparative proteome analyses were performed and the Venn diagram was generated using orthovenn (Fig. 7 and Table 6).

Discussion

S. maltophilia is one of the leading causes of nosocomial infections worldwide. It has been reported by [40] that there has been an increasing resistance among *S. maltophilia* along with other microbes. Thus, there is a need to devise proper surveillance of antibiotic resistance among the clinical strains. This study has focused on the whole genome sequencing of *S. maltophilia* in order to identify the genes which are responsible for antibiotic resistance. Initially, 25 samples were taken but as reported by [41] that the whole genome sequencing is a costly process, so, out of the 25 isolates, only two isolates were selected which were properly identified through physical and biochemical testing.

Whole genome sequencing of the strains was carried out which determined that the strains comprised of 4,668,850 base pairs genome with about 65.8% genomic content. The presence of genomic content is significant in investigating antimicrobial resistance profiling as it has been pointed out by [42] that GC content confers stability to the genome of the bacteria thus the antibiotic involving degradation of the genome may become less effective in inhibiting those isolates with higher GC content. The results obtained in this study are in agreement with the findings reported by [43] who also carried out whole genome sequencing of *S. maltophilia* and found that its genome comprised of 4.75 Mb base pairs along with 66.5% of the total GC content. The genome of metal-resistant *S. maltophilia* isolated from the environment by [44] also comprised of similar length of base pairs i.e., 4,968,240, and have a GC content of almost 66% of the total genome. Thus, it can be understood that the GC content of *S. maltophilia* is more than 50% and it provides genome stability.

Furthermore, the comprehensive antibiotic resistance database (CARD) shows both strains contain four resistance genes *adeF*, *qacG*, *adeF*, and *smeR*. The resistance genes determination by [45] also reported that the *S. maltophilia* strains possess *adeF* gene that confers resistance to chloramphenicol and trimethoprim. The presence of this gene was in line with the presence of chloramphenicol resistance in the selected strains. Moreover, the study conducted by [46] also reported the simultaneous existence of *adeF* and *smeR* genes that are contributing to the antibiotic resistance. The gene *OqxB* is also found to be present in the selected strains. The presence of this gene is found to be associated with the production of the efflux pumps that are involved in the active excretion of the drug out of the bacteria cell. The simultaneous occurrence of the two genes i.e., *OqxB* and *BlaL1* are also reported by [47]. The whole genome sequencing conducted by [48] showed that the isolates that are susceptible to levofloxacin possess *qnrB* gene, while this study did not investigate the presence of *qnrB* gene in the selected strains. Thus, these results when comparing with the other studies showed that *S.*



Fig. 6. : Clusters group protein of isolates.

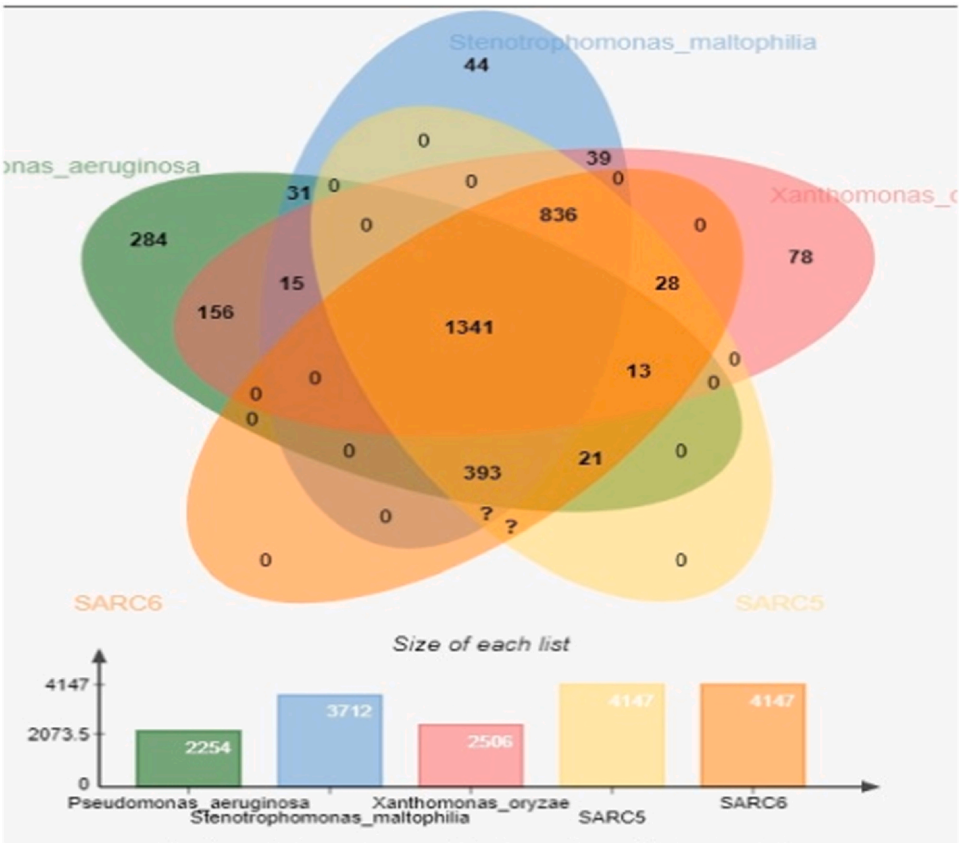


Fig. 7. : Venn diagram of the *Stenotrophomonas* genome showing the number of orthologous genes.

Table 6

Summary of Venn diagram showing the number of orthologous genes.

Species	Protein	Clusters	Singletons
<i>Pseudomonas aeruginosa</i>	5718	2254	2806
<i>Stenotrophomonas maltophilia</i>	4543	3712	721
<i>Xanthomonas oryzae</i>	3999	2506	931
<i>Stenotrophomonas maltophilia</i>	4179	4147	9
<i>Stenotrophomonas maltophilia</i>	4179	4147	9

maltophilia has become an increasing threat to public health as it is acquiring antibiotic resistance genes. The simultaneous occurrence of the antibiotic resistance genes could result in the development of multidrug resistance and thus there would be fewer treatment options available to treat the infections caused by it.

Conclusion

S. maltophilia is a misclassified species present in different matrices. The whole genome sequence analysis of both SARC-5 and SARC-6 revealed the presence of four resistance genes *adeF*, *qacG*, *adeF*, and *smeR*. Moreover, *OqxB* gene is also present in *S. maltophilia*, which has the potential to subsequently excrete the drug from the bacterial cell. It has been concluded that *S. maltophilia* has antibiotic resistance genes which could develop multidrug resistance in the human hosts. Furthermore, this investigation laid the groundwork for future research into the mechanisms of *S. maltophilia* resistance in Pakistan and elsewhere, which would be extremely beneficial in clinical studies.

Ethical approval and consent to publish

This article does not contain any studies with human and animal participants.

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CRediT authorship contribution statement

Conceptualization, S.G, R.R, A.A.A, M.A; Data curation, S.G, R.R, M.E.A and Y.K; Investigation, R.A, Y.K, L.J; Methodology, S.S, R.R, R.A, Y.K; Project administration, A.A¹, A.A², S.G, R.A, S.M.B.A, M.A; Software, S.G; Supervision, A.A¹, A.A², S.G, Y.K, A.A.A, M.A; Validation, W.A, S.S, M.E.A; Writing – original draft, S.S, W.A and R.A; Writing – review & editing, R.A, A.A.A, S.M.B.A. Funding support, A.A¹, A.A² and S.M.B.A. All authors have read and agreed to the published version of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Informed consent statement

Not applicable.

Appendix

Not Applicable.

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